Tutorial 1 Recap on Number Systems and Digital Systems
ES 336 Computer Organization and Architecture
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Problem 1 Perform the following number conversions:

A. hex 0x25B9D2 to binary

Solution Each hexadecimal digit is represented using 4 binary bits:

$$2 = 0010$$
$$5 = 0101$$
$$B = 1011$$
$$9 = 1001$$
$$D = 1101$$
$$2 = 0010$$

$$0x25B9D2 = 0010\,0101\,1011\,1001\,1101\,0010_2$$

B. binary 1010111001001001 to hexadecimal

Solution Grouping into 4-bit nibbles:

$$1010\,1110\,0100\,1001$$

$$1010 = A, \quad 1110 = E, \quad 0100 = 4, \quad 1001 = 9$$

$$1010111001001001_2 = 0xAE49$$

C. hex 0x36 to decimal

Solution

$$0x36 = (3 \times 16^1) + (6 \times 16^0) = 48 + 6 = 54$$

D. decimal 771 to hex

Solution

$$771 \div 16 = 48 \text{ remainder } 3$$
$$48 \div 16 = 3 \text{ remainder } 0$$
$$3 \div 16 = 0 \text{ remainder } 3$$

Reading remainders from bottom to top:

$$771_{10} = 303_{16}$$

Problem 2 Represent the decimal number +23 in:

A. Unsigned binary

B. Signed magnitude

C. One's complement

D. Two's complement (using 8 bits)

Solution
First, convert +23 to binary:

$$23 = 16 + 4 + 2 + 1$$

$$23_{10} = 00010111_2$$

A. **Unsigned Binary**

$$00010111$$

B. **Signed Magnitude**

The number is positive, so the sign bit is 0:

$$00010111$$

C. **One's Complement**

Positive numbers remain unchanged:

$$00010111$$

D. **Two's Complement (8-bit)**

Positive numbers remain unchanged:

$$00010111$$

Problem 3 Convert the 8-bit binary number "10010110" to decimal, assuming they are:

A. Unsigned

B. Signed magnitude

C. One's complement

D. Two's complement

Solution

A. **Unsigned Representation**

$$\begin{aligned}10010110_2 &= (1 \times 128) + (0 \times 64) + (0 \times 32) + (1 \times 16) \\ &\quad + (0 \times 8) + (1 \times 4) + (1 \times 2) + (0 \times 1) \\ &= 128 + 16 + 4 + 2 \\ &= 150\end{aligned}$$

B. **Signed Magnitude**

The most significant bit is 1, indicating a negative number. The remaining bits represent the magnitude:

$$0010110_2 = 16 + 4 + 2 = 22$$

$$\text{Decimal value} = -22$$

C. **One's Complement**

The number is negative since the most significant bit is 1. Taking one's complement:

$$10010110_2 \rightarrow 01101001_2$$

$$01101001_2 = 64 + 32 + 8 + 1 = 105$$

$$\text{Decimal value} = -105$$

D. **Two's Complement**

Taking two's complement:

$$\begin{aligned}10010110_2 &\rightarrow 01101001_2 \\ &\rightarrow 01101010_2\end{aligned}$$

$$01101010_2 = 64 + 32 + 8 + 2 = 106$$

$$\text{Decimal value} = -106$$

Problem 4 Write the binary representation of -45 using

A. Signed magnitude

B. One's complement

C. Two's complement (using 8 bits)

Solution

First, convert +45 to binary:

$$45 = 32 + 8 + 4 + 1$$

$$45_{10} = 00101101_2$$

A. **Signed Magnitude**

   Sign bit = 1, magnitude = 0101101:

$$10101101$$

B. **One's Complement**

   Inverting all bits:

$$00101101_2 \rightarrow 11010010_2$$

C. **Two's Complement (8-bit)**

   Adding 1 to the one's complement:

$$11010010_2 + 1 = 11010011_2$$

Problem 5 Perform the following conversions:

A. Convert the decimal number 13.625 into IEEE-754 single-precision floating-point representation.

   Solution

   Step 1: Convert integer and fractional parts to binary

$$13_{10} = 1101_2$$

$$0.625_{10} = 0.101_2$$

$$13.625_{10} = 1101.101_2$$

   Step 2: Normalize the number

$$1101.101_2 = 1.101101 \times 2^3$$

   Step 3: Determine sign, exponent, and mantissa

   - Sign bit = 0 (positive number)
   - Exponent = $3 + 127 = 130 = 10000010_2$
   - Mantissa = bits after the binary point: 10110100000000000000000

Step 4: IEEE-754 representation

$$\boxed{0\ 10000010\ 10110100000000000000000}$$

B. Convert the binary floating-point number -1.1011 $\times$ 2^3 into IEEE-754 format.

Solution

Step 1: Identify sign

$$\text{Sign bit} = 1$$

Step 2: Normalize (already normalized)

$$1.1011 \times 2^3$$

Step 3: Calculate exponent

$$3 + 127 = 130 = 10000010_2$$

Step 4: Determine mantissa

Mantissa = 10110000000000000000000

Step 5: IEEE-754 representation

$$\boxed{1\ 10000010\ 10110000000000000000000}$$

C. Decode the following IEEE-754 single-precision number into decimal:
0 10000010 01100000000000000000000

Solution

Step 1: Sign bit

$$\text{Sign} = 0 \Rightarrow \text{Positive number}$$

Step 2: Exponent

$$10000010_2 = 130_{10}$$

$$\text{Actual exponent} = 130 - 127 = 3$$

Step 3: Mantissa

$$1.011_2 = 1 + 0.25 + 0.125 = 1.375$$

Step 4: Final value

$$1.375 \times 2^3 = 11$$

Decoded decimal value:

$$\boxed{11_{10}}$$

Problem 6 State De-Morgan's law and prove it using truth tables.

Solution

De Morgan's Laws state that:

1. The complement of a sum is equal to the product of the complements:

$$(A + B)' = A' \cdot B'$$

2. The complement of a product is equal to the sum of the complements:

$$(A \cdot B)' = A' + B'$$

Proof Using Truth Tables

Law 1: $(A + B)' = A' \cdot B'$

A	B	$A + B$	$(A + B)'$	$A'B'$
0	0	0	1	1
0	1	1	0	0
1	0	1	0	0
1	1	1	0	0

Since the columns $(A + B)'$ and $A'B'$ are identical, the law is verified.

Law 2: $(A \cdot B)' = A' + B'$

A	B	AB	$(AB)'$	$A' + B'$
0	0	0	1	1
0	1	0	1	1
1	0	0	1	1
1	1	1	0	0

Since the columns $(AB)'$ and $A' + B'$ are identical, the law is verified.

Conclusion

Thus, both De Morgan's laws are proven using truth tables.

Problem 7 Simplify the following boolean expressions using De-Morgan's Law

1. $F = \overline{A + B + C}$

Solution

$$\begin{aligned}(A+B+C)' &= \big((A+B)+C\big)' \\ &= (A+B)'\cdot C' \\ &= (A'\cdot B')\cdot C' \\ &= A'\cdot B'\cdot C'\end{aligned}$$

$$\boxed{F = A'B'C'}$$

2. $F = \overline{(A+B)(C+D)}$

   Solution

$$\begin{aligned}(A+B)' &= A'\cdot B' \\ (C+D)' &= C'\cdot D'\end{aligned}$$

$$\begin{aligned}F &= (A+B)' + (C+D)' \\ &= A'\cdot B' + C'\cdot D'\end{aligned}$$

$$\boxed{F = A'B' + C'D'}$$

3. $F = \overline{AB + \overline{C}D}$

   Solution

$$\begin{aligned}F &= (AB + C'D)' \\ &= (AB)'\cdot(C'D)' \\ &= (A'+B')\cdot(C+D')\end{aligned}$$

$$\boxed{F = (A'+B')\cdot(C+D')}$$

Problem 8 Minimize the following boolean expressions using K-Map

1. $F(A,B) = \sum m(1,3)$

   Solution B

2. $F(A,B,C) = \sum m(1,3,5,7)$

   Solution C

3. $F(A,B,C) = \prod M(0,2,5,7)$

   Solution $(A'+C')\cdot(A+C)$

4. $F = \overline{A}\,\overline{B}C + \overline{A}BC + AB\overline{C}$

Solution $A'C + ABC'$

5. $F = AB\overline{C}D + A\overline{B}CD + ABCD + \overline{A}BCD$

   Solution $ABD + ACD + BCD$

6. $F = A\overline{B} + B\overline{C} + AC$

   Solution $A + BC'$